

Lecture 12: Meteorites, Asteroids, Minor Planets & Comets

Planetary Systems

Tim Lichtenberg
Kapteyn Astronomical Institute
University of Groningen

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Learning Objectives

By the end of this lecture, you will be able to:

- Classify meteorites and derive the Pb-Pb isochron age of calcium-aluminium-rich inclusions (CAIs)
- Describe the asteroid-belt and trans-Neptunian populations and their dynamics
- Explain the origin, anatomy and volatile composition of comets, and read small bodies as fossils of planet formation

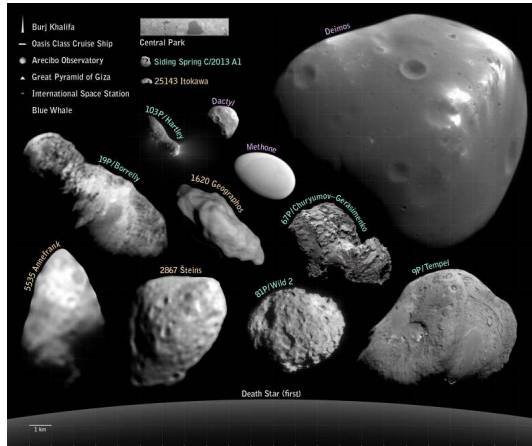
Recap: Where We Are in the Course

- Lectures 9–11 surveyed the major planets across the rocky and giant classes
- Today: the leftovers; asteroids, meteorites, KBOs, comets, and the Oort cloud
- Small bodies act as formation fossils preserving disk chemistry and migration history

Today: What do the leftovers reveal about the origin of planetary systems?



1. Overview: The Reservoirs



Small bodies visited by spacecraft, scaled against terrestrial landmarks. Credit: NASA/JPL-Caltech, public domain.

Three reservoirs hold them: the main belt ($2.1\text{--}3.3\text{ AU}$, $\sim 4 \times 10^{-4} M_{\oplus}$), the Kuiper Belt, and the Oort cloud ($2,000\text{--}50,000\text{ AU}$, inferred, never directly observed).

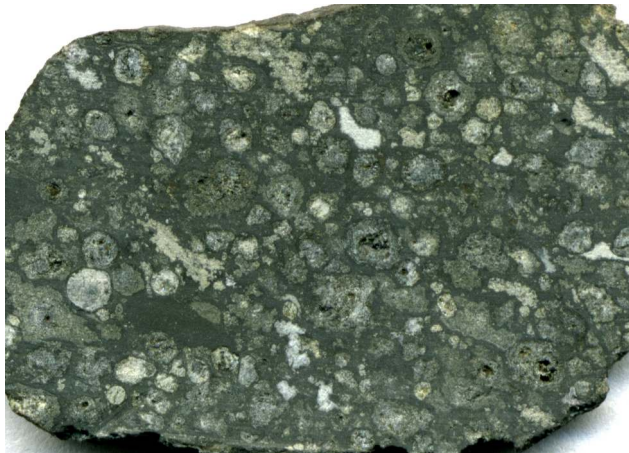


2. Meteorite Taxonomy

Allende meteorite (CV3 chondrite, fell 1969). J. St. John CC BY 2.0

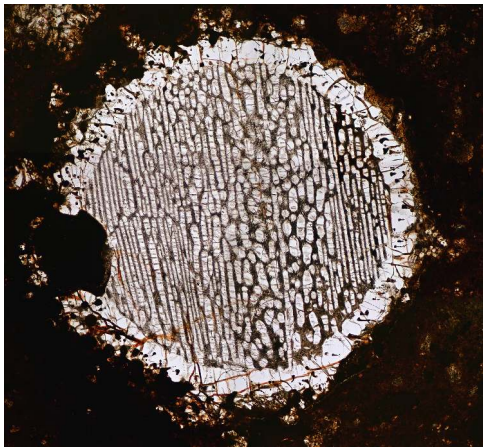
Why Meteorites Matter

- **Laboratory access:** ~70,000 catalogued specimens; the primitive ones are older than the oldest terrestrial zircons (~ 4.4 Ga)
- **Presolar material:** primitive chondrites contain grains that condensed around other stars, older than the Sun
- **Anchor age:** CAI crystallization age of 4567.30 ± 0.16 Myr anchors solar system history



Allende slab: chondrules and bright CAIs in a dark matrix. Credit: James St. John, CC BY 2.0.

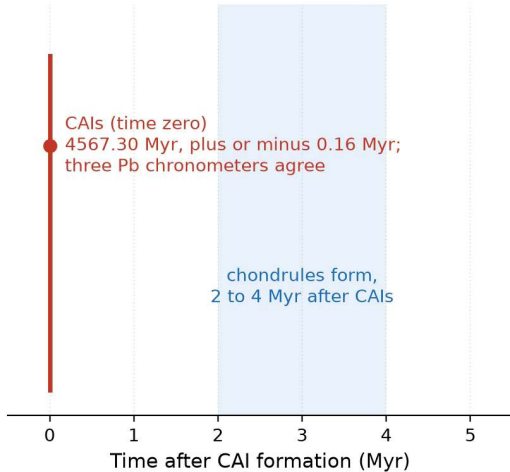
Chondrites never melted whole and keep their mm-scale chondrules; achondrites, irons and stony irons come from parent bodies that short-lived ^{26}Al melted and differentiated: irons are cores, achondrites crusts and mantles.



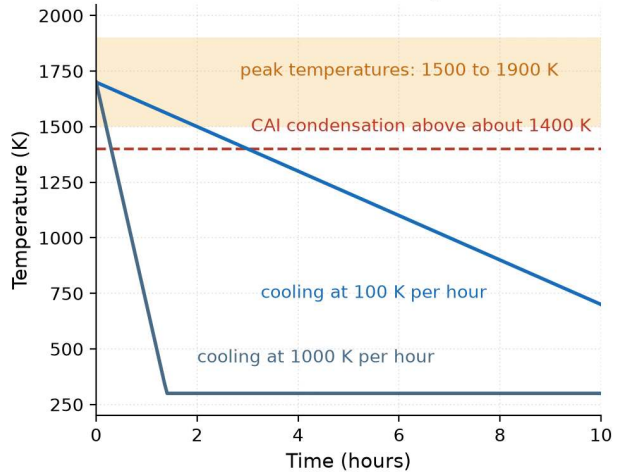
Cross-polarised thin section of NWA 5930: each chondrule is a mm-scale bead of olivine and pyroxene. Credit: H. Raab, CC BY-SA 3.0 (Wikimedia Commons).

Each chondrule froze from a fully molten droplet in seconds to hours from a peak of 1500–1900 K; how chondrules form remains an unsolved problem.

(a) The oldest solids

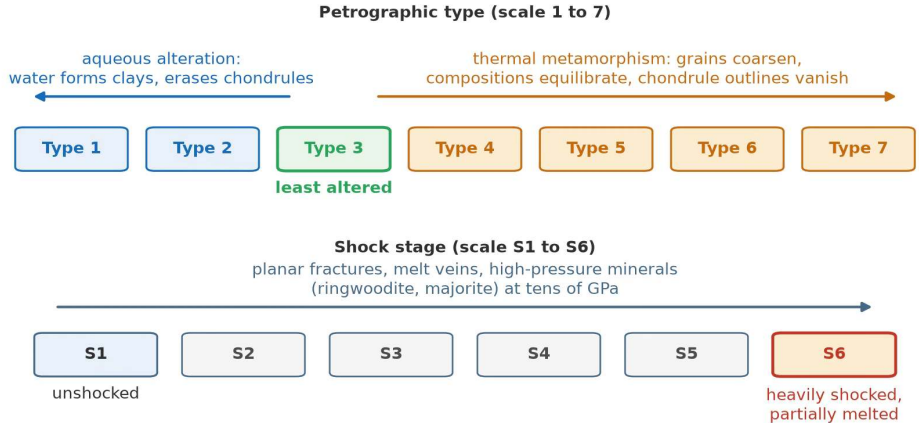


(b) A chondrule's heating event



(a) CAIs at 4567.30 Myr define $t = 0$ and chondrules formed 2 to 4 Myr later; (b) a chondrule heating event from a 1700 K peak, cooling at 100 and 1000 K per hour. Course figure.

Two scales that record what happened on the parent body

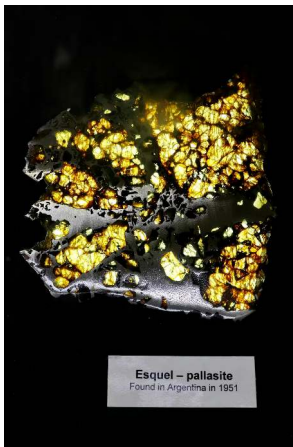


Petrographic types 1 to 7 (aqueous alteration to thermal metamorphism) and shock stages S1 to S6 record what the parent body did. Course schematic.



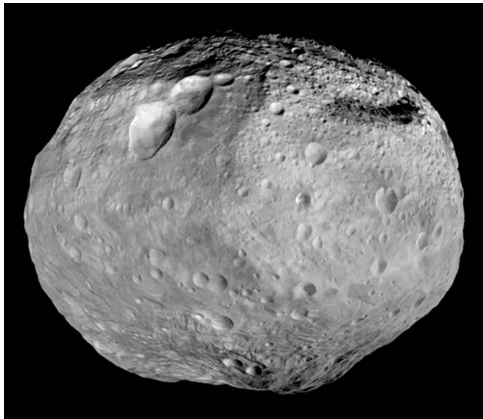
Widmanstätten pattern in an iron meteorite: a kamacite and taenite Fe-Ni lattice. Credit: H. Raab, CC BY-SA 3.0 (Wikimedia Commons).

The lattice grows only at cooling rates $\leq 100 \text{ K Myr}^{-1}$, inside the metallic core of a body tens of km across; its spacing gives the original parent-body radius.



Esquel pallasite: yellow-green olivine in silvery Fe-Ni metal. Credit: UCLA Meteorite Gallery, CC BY-SA 4.0 (Wikimedia Commons).

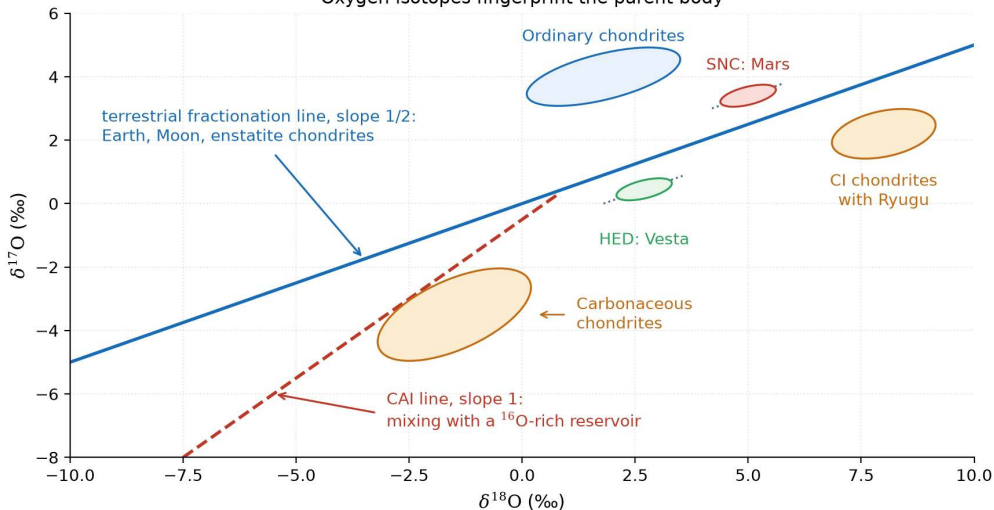
Likely a sample of a planetesimal's core-mantle boundary: molten metal mixed into the olivine of the mantle, possibly during a giant impact.



Vesta seen by Dawn (2011–2012), the parent body of the HED meteorites. Credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA, public domain.

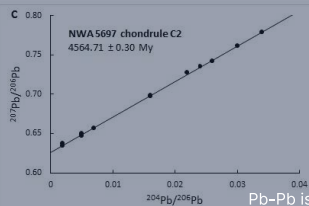
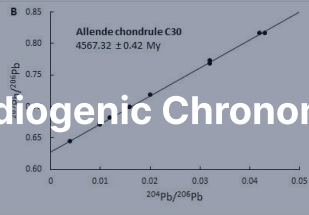
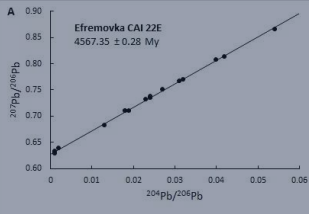
HEDs (howardites, eucrites, diogenites, ~ 6% of falls) match Vesta in mineralogy and oxygen isotopes and were ejected from the Rheasilvia basin; lunar meteorites match the Apollo samples, Martian ones carry trapped Mars atmosphere.

Oxygen isotopes fingerprint the parent body



Oxygen three-isotope plot: the terrestrial line of slope 1/2, the CAI slope-1 mixing line towards pure ^{16}O , and the chondrite, HED and SNC fields. Course figure.

3. Radiogenic Chronometers



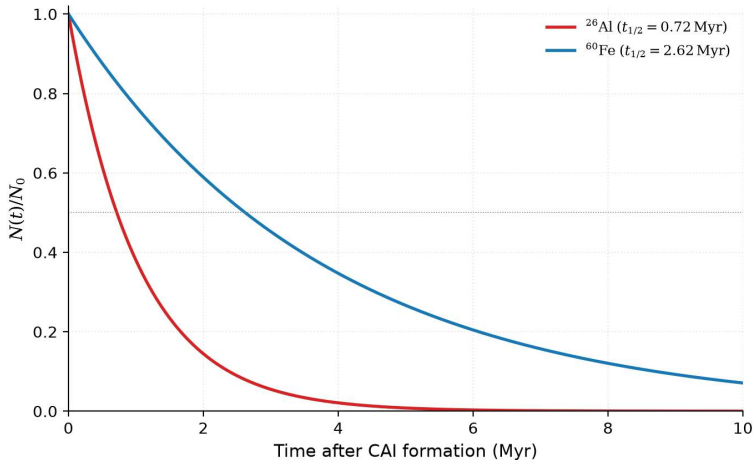
Pb-Pb isochrons for CAI 22E and chondrules. Connelly et al. (2012)

Decay Clocks and Early Solar System Timeline

Radioactive decay $\frac{dN}{dt} = -\lambda N$ ($t_{1/2} = \ln 2 / \lambda$) provides absolute and relative chronometers:

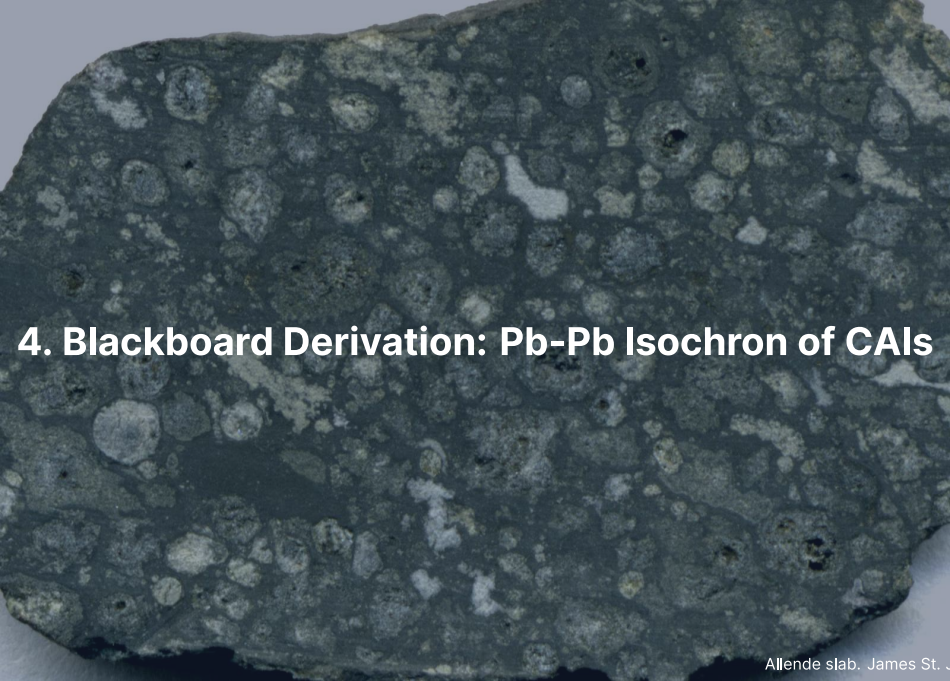
- **Long-lived:** ^{238}U , ^{235}U yield absolute ages; CAIs condense at $t_0 = 4567.30 \pm 0.16$ Myr
- **Short-lived (extinct):** ^{26}Al (0.717 Myr), ^{182}Hf (8.9 Myr) yield relative ages
- **Early chronology:** differentiation at 0.5–4 Myr, chondrules at 1–4 Myr after CAIs

Long-lived chronometers anchor absolute time; extinct nuclides resolve events to < 1 Myr.



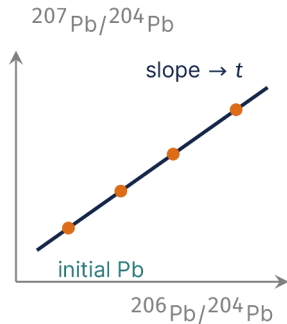
Abundance decay of ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (2.62 Myr) over the first 10 Myr after CAI formation. Course figure.

Their rapid decay powered early planetesimal melting and differentiation.

A photograph of a dark, irregularly shaped meteorite slab, likely the Allende meteorite. The surface is covered with numerous small, light-colored, rounded inclusions (chondrules) set in a darker matrix. The text "4. Blackboard Derivation: Pb-Pb Isochron of CAIs" is overlaid in white on the left side of the image.

4. Blackboard Derivation: Pb-Pb Isochron of CAIs

Goal and Strategy



Goal: the age of the oldest solids from two parallel decay chains of one element.

- $^{238}\text{U} \rightarrow ^{206}\text{Pb}$ (4.4683 Gyr) and $^{235}\text{U} \rightarrow ^{207}\text{Pb}$ (0.7038 Gyr); ^{204}Pb is the stable reference
- Daughter since closure:
 $^{206}\text{Pb}^* = ^{238}\text{U} [e^{\lambda_{238}t} - 1]$, same for 207; add the initial Pb
- Divide the 207 line by the 206 line so the U/Pb of the rock drops out

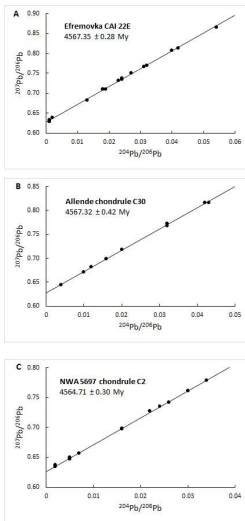
To the board.

Result: The Pb-Pb Isochron

Cogenetic samples with unknown initial Pb fall on a line in ($^{206}\text{Pb}/^{204}\text{Pb}$, $^{207}\text{Pb}/^{204}\text{Pb}$) whose slope depends only on t :

$$\frac{^{207}\text{Pb}^*}{^{206}\text{Pb}^*} = \frac{^{235}\text{U}}{^{238}\text{U}} \frac{e^{\lambda_{235}t} - 1}{e^{\lambda_{238}t} - 1}, \quad \frac{^{235}\text{U}}{^{238}\text{U}} = \frac{1}{137.8}$$

The slope is independent of the initial Pb and of the U/Pb ratio: the age needs no assumption about the starting composition.



Pb-Pb isochrons for CAI 22E (Efremovka) and chondrules in the inverse form: the intercept of the mineral-fraction line at zero ^{204}Pb gives the radiogenic $^{207}\text{Pb}/^{206}\text{Pb}$ and so the age t , independent of the initial Pb. From Connelly et al. (2012).

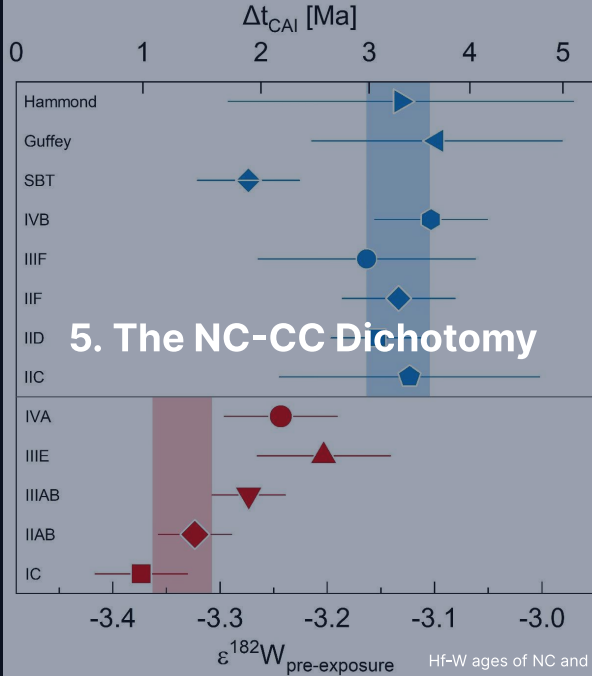
Step 5: Numerical Result

Connelly et al. (2012), CAIs from Efremovka (CV3), one from Allende, after acid leaching:

$$t_{\text{CAI}} = 4567.30 \pm 0.16 \text{ Myr}$$

- Independent samples and labs agree to the same precision (Amelin et al. 2010)
- Chondrules from Allende and NWA 5697: a few Myr younger
- The *zero point* of the solar system clock

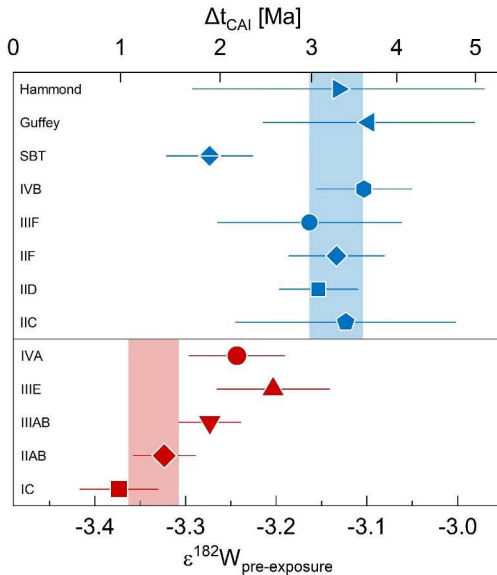
Two parallel decay chains make the problem self-calibrating.



Two Isotopic Reservoirs

Nucleosynthetic anomalies in Ti, Cr, Mo, Ru and Ca divide solar system materials into two isolated reservoirs:

- **NC reservoir (non-carbonaceous):** ordinary and enstatite chondrites, HEDs, Earth, and Mars
- **CC reservoir (carbonaceous):** carbonaceous chondrites, several iron groups, and outer bodies
- **Prolonged isolation:** the two reservoirs did not mix isotopically for ~ 3–4 Myr after CAIs



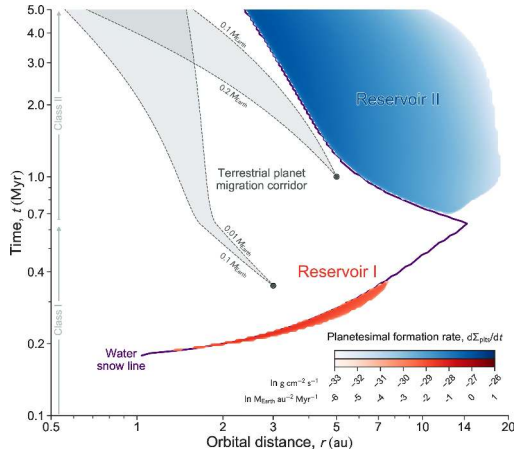
Hf-W ages of NC (red) and CC (blue) iron meteorites: $\epsilon^{182}\text{W}$ below, the age it gives above. Spitzer et al. (2021).

NC irons (-3.37 to -3.20) formed cores 1.0–2.6 Myr after CAIs, CC irons (-3.16 to -3.10) at 3.0–3.6 Myr. NC cores form ~ 2 Myr before CC cores; the ungrouped SBT is ~ 1 Myr earlier.

Three Live Interpretations

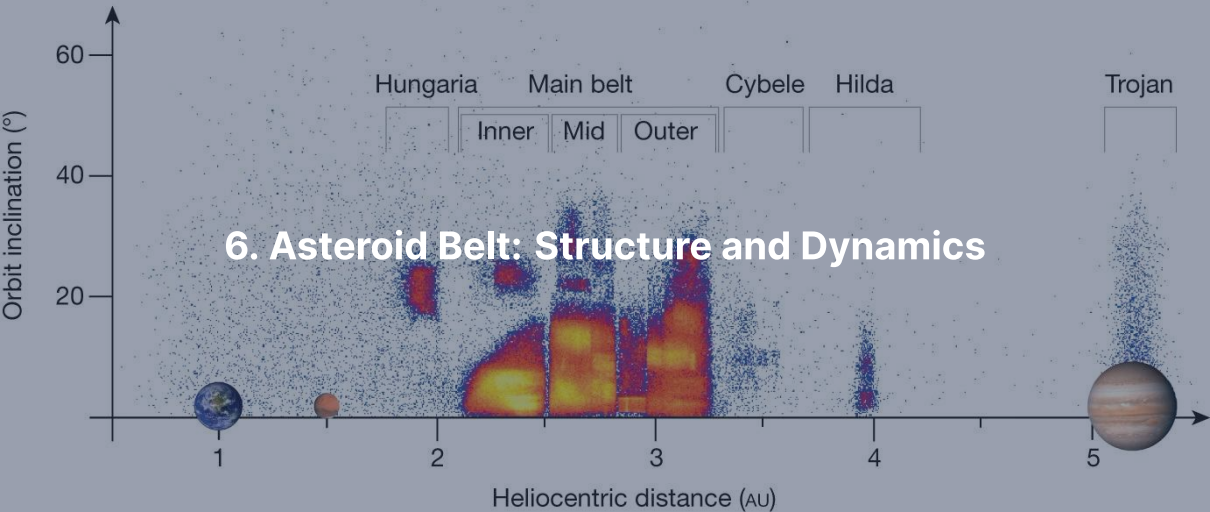
Three competing mechanisms explain the NC-CC dichotomy from identical cosmochemical data:

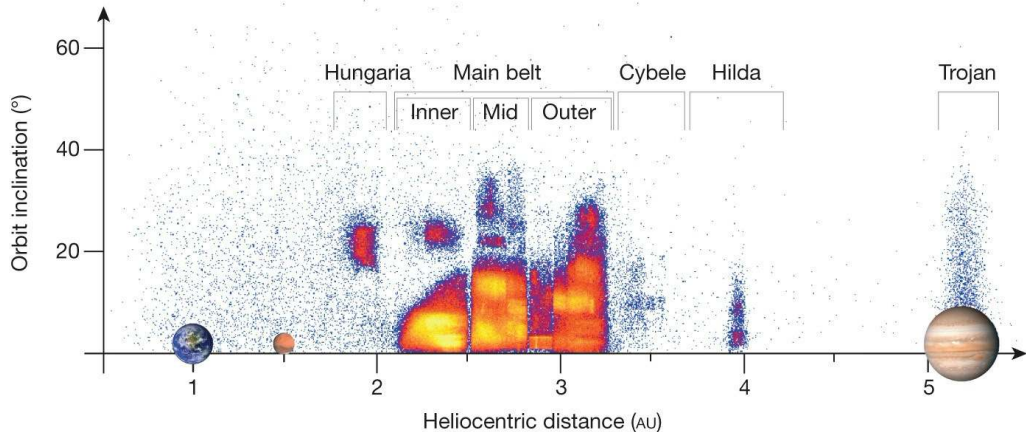
- **Jupiter barrier:** early proto-Jupiter ($\lesssim 1$ Myr) opens a gap, blocking inward pebble drift (Kruijer et al. 2017)
- **Snow line:** disk thermal evolution produces two successive planetesimal formation epochs (Lichtenberg et al. 2021)
- **Temporal change:** inward disk drift alters feeding composition over time (Schiller et al. 2018)



Disk simulation in time against orbital distance with two planetesimal formation reservoirs.
Lichtenberg et al. (2021).

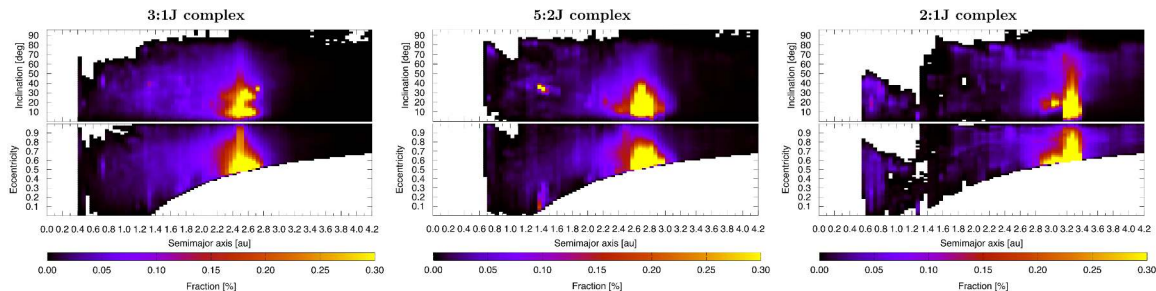
Reservoir I (red), early and dry in the inner disk, is NC; Reservoir II (blue), later and ice-rich in the outer disk, is CC: the bifurcation arises from disk evolution itself.





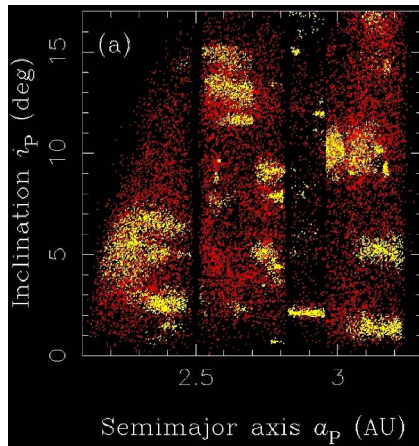
Asteroid orbits in inclination against heliocentric distance: Hungarias, the main belt at 2.1–3.3 AU, Hildas, and the Trojans at 5.2 AU. DeMeo & Carry (2014).

About 1.4 million catalogued bodies, $\sim 10^6$ above 1 km, a total mass of $\sim 4 \times 10^{-4} M_{\oplus}$ (Ceres $\approx 40\%$); the vertical gaps are the Kirkwood resonances with Jupiter.



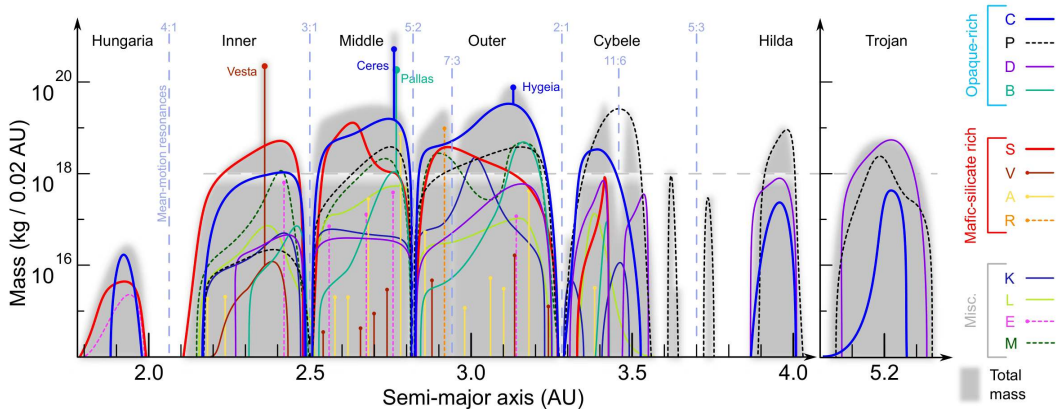
Orbits of the near-Earth objects that leave the main belt through the 3:1 (2.50 AU), 5:2 (2.82) and 2:1 (3.27) resonances with Jupiter. Granvik et al. (2018).

Periodic Jovian kicks add coherently and pump eccentricities to planet-crossing in 10^4 – 10^6 yr: the gaps are not empty but leaky, the dynamical sinks that feed the near-Earth population.



Asteroid families as clusters in proper inclination against proper semimajor axis: Themis, Eos, Koronis, Eunomia, Vesta, Flora. Nesvorný et al. (2015).

Each family is the debris of a single disruption event; its spread gives the original parent body's size and age.



Mass of the main belt by spectral class against semimajor axis. DeMeo & Carry (2014).

S-types (~ 15% albedo, ordinary-chondrite analogues) fill the inner belt, **C-types** (~ 4% albedo, volatile-rich) ~ 75% of the outer belt, **M- and V-types** are cores and Vesta's basaltic crust (2.4 AU). Radial sorting preserves the disk's primordial volatile gradient.

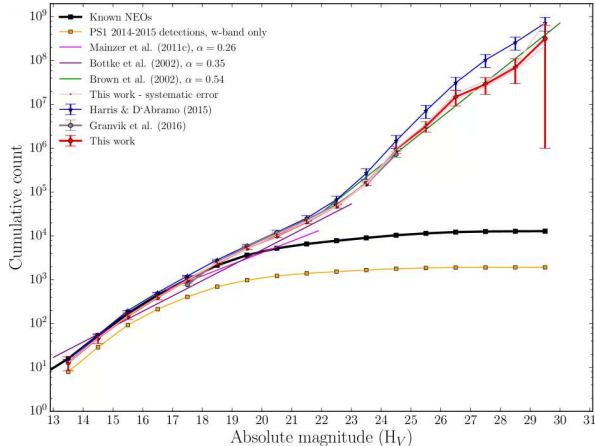
A photograph of a meteor streaking across a twilight sky. The meteor is a bright, white, elongated trail of smoke and dust, starting from the upper left and moving towards the lower right. Below the sky, a dark silhouette of a village with snow-covered roofs and bare trees is visible against the horizon. The overall scene is dimly lit, with the sky transitioning from a deep blue at the top to a lighter, hazy glow near the horizon.

7. NEAs, Yarkovsky/YORP, and Planetary Defence

Near-Earth Asteroids and Resupply Channels

NEAs ($q < 1.3$ AU, residence time ~ 10 Myr) require steady-state resupply from the main belt:

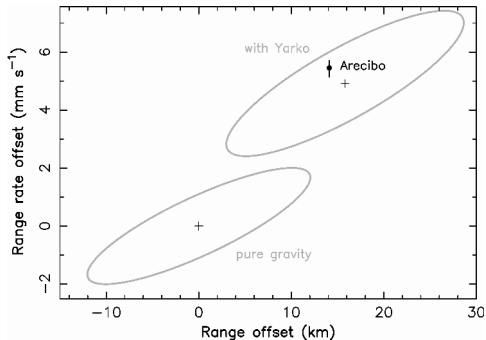
- **Resonances:** Kirkwood (3:1, 5:2) and ν_6 secular resonance pump eccentricity
- **Yarkovsky drift:** thermal recoil drives slow semimajor axis drift,
$$\frac{da}{dt} \propto \frac{1}{D \rho \sqrt{a}}$$
- **Hazards:** $\sim 2,500$ Potentially Hazardous Asteroids ($D > 140$ m, MOID < 0.05 AU)



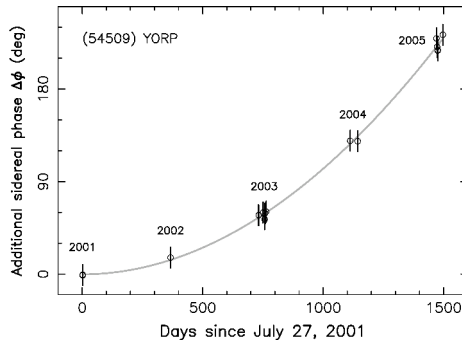
Cumulative size-frequency distribution of near-Earth objects against absolute magnitude H_V , a power law over five orders of magnitude. Credit: Schunova et al. (2017), Fig. 9.

Impact rates: 10 m per decade; 50 m (Tunguska) per 10^3 yr; 1 km per 5×10^5 yr; 10 km (Chicxulub) per 10^8 yr.

Yarkovsky and YORP Detected on Real NEAs

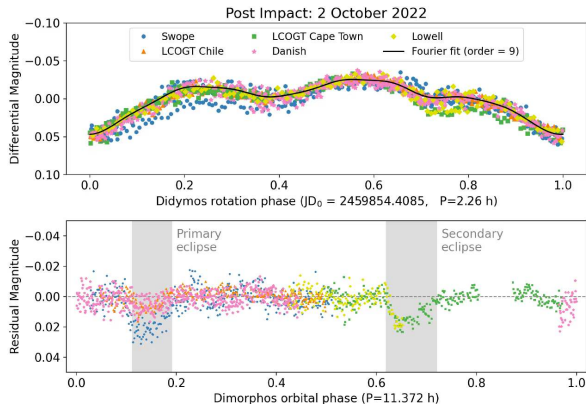


Yarkovsky: (6489) Golevka, Arecibo radar



YORP: (54509) YORP rotational acceleration

YORP is the same thermal asymmetry acting on *spin*: it can spin small asteroids to fission or stop them.



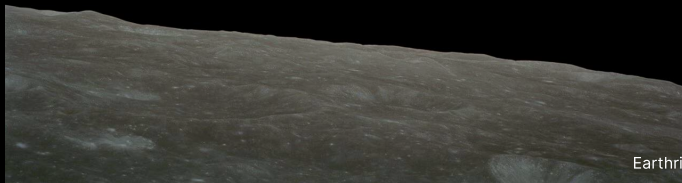
Light curves of Didymos (top, 2.26 h rotation) and Dimorphos (bottom, its new 11.372 h orbit) six days after the DART impact. Thomas et al. (2023), Fig. 3.

The orbital period fell by 33.0 ± 1.0 minutes (3σ) and the momentum enhancement $\beta \approx 3.6$ shows that ejecta recoil exceeds the direct push; Hera (ESA, launched 2024) measures the crater and β .

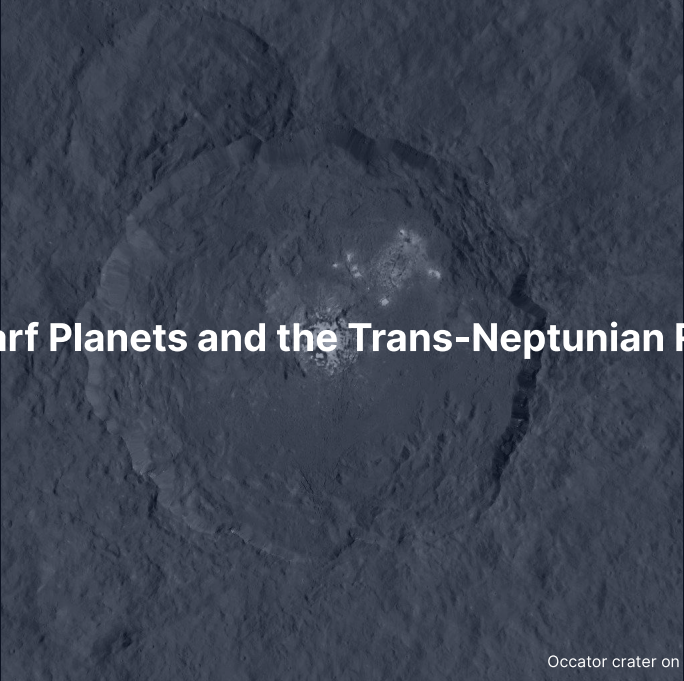
Break



End of Part 1: Meteorites & Inner Small Bodies
Part 2: Trans-Neptunian Region, Comets, Missions

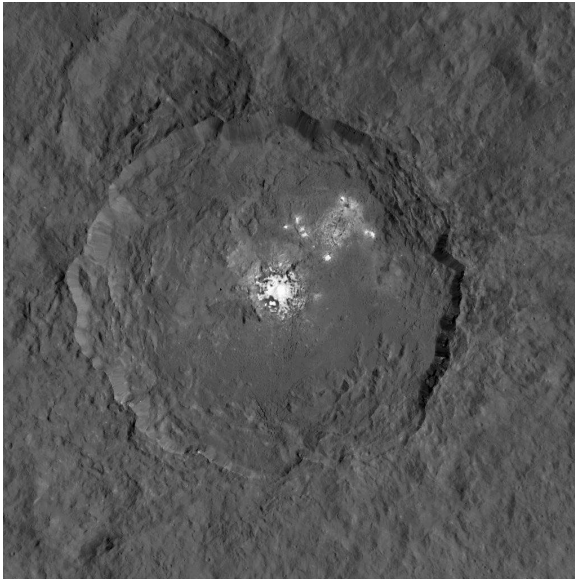


Earthrise, Apollo 8 (1968). Credit: NASA

A grayscale image of the Occator crater on the dwarf planet Ceres. The crater is a large, roughly circular depression with a prominent, bright, and irregularly shaped central feature. The surrounding terrain is heavily textured with smaller craters and ridges. The image is set against a dark blue background.

8. Dwarf Planets and the Trans-Neptunian Region

Occator crater on Ceres, Dawn. NASA/JPL-Caltech



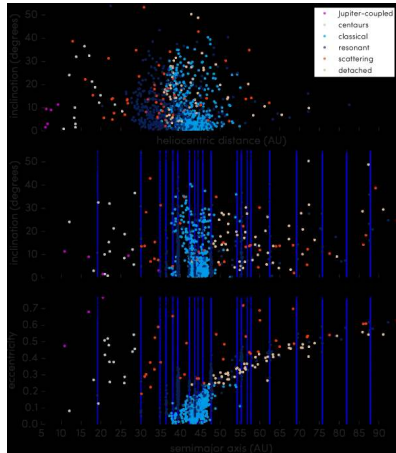
Occator crater on Ceres, Dawn (2015–2018): bright deposits of hydrated Na_2CO_3 and NH_4Cl from erupted brines. Credit: NASA/JPL-Caltech.

Ceres ($R = 470$ km, mass $\sim 1\%$ of the Moon) has, or had, a partially liquid water layer at depth, and may be a captured outer-solar-system body (Nice Model: a dynamical instability among the giant planets).

Trans-Neptunian Region: Five Sub-Populations

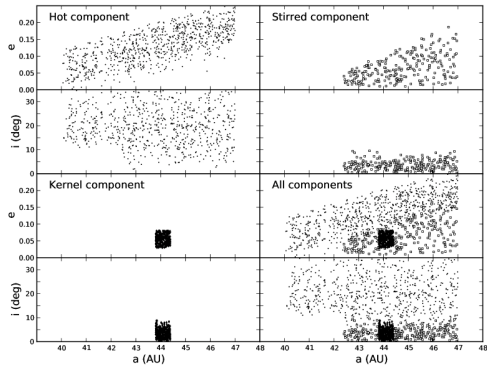
The orbits preserve the gravitational footprint of giant-planet migration:

- **Classicals ($a = 42\text{--}47$ AU):** dynamically unperturbed cold classicals ($i \lesssim 5^\circ$) and scattered hot classicals ($i \leq 30^\circ$)
- **Resonant (e.g. 3:2 plutinos):** bodies swept into mean-motion resonances during Neptune's outward migration
- **Scattered disk and detached:** Neptune-scattered bodies ($q \sim 30$ AU) and decoupled detached objects ($q > 40$ AU)



1142 characterised TNOs from OSSOS in orbital space; the vertical lines mark Neptune's resonances. Bannister et al. (2018).

The dense low-inclination cluster at 42–47 AU is the cold classical belt; the architecture is a fossil of giant-planet migration.



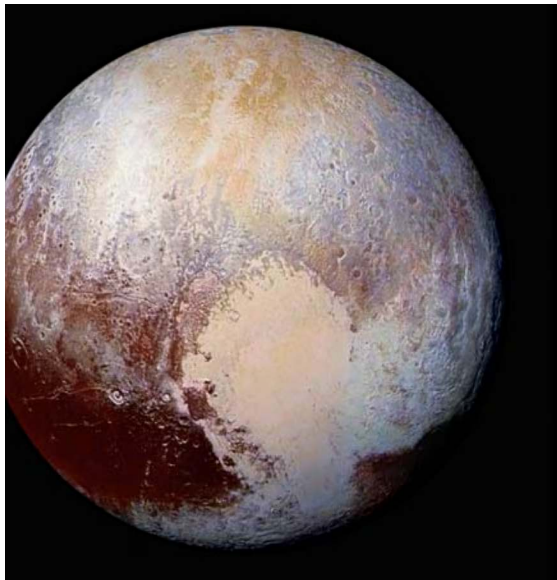
The three components of the CFEPS-L7 model of the classical Kuiper belt in e and i against a : the hot component, the stirred component, and the dense low-inclination kernel near 44 AU.

Petit et al. (2011).

Kernel and hot component share the same a but different (e, i) : the cold classicals were never strongly perturbed by Neptune, the hot classicals were excited by the giant-planet instability.

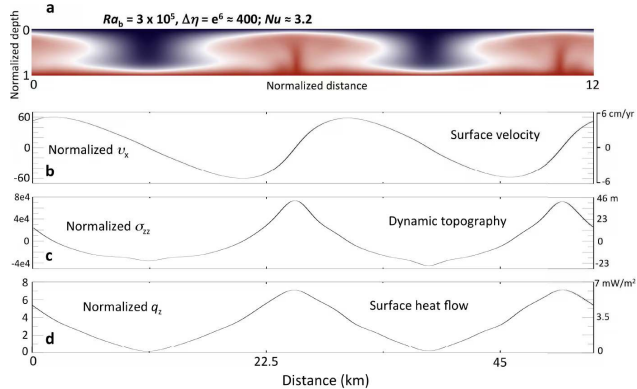


9. Pluto, Charon, Arrokoth



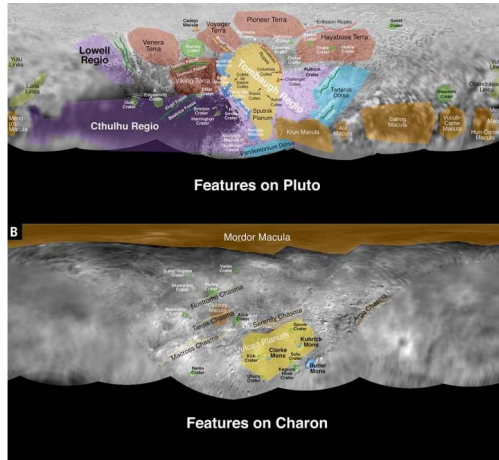
True-colour Pluto from New Horizons Ralph (2015), with Tombaugh Regio and Sputnik Planitia. Credit: NASA/JHUAPL/SwRI.

With $R = 1187$ km and $\rho = 1860$ kg m⁻³: water-ice bedrock under volatile frosts, a convecting N₂-ice basin younger than 10 Myr, a ~ 1 Pa N₂ atmosphere and a probable subsurface ocean.



Numerical model of solid-state convection in Sputnik Planitia at $Ra_b \sim 3 \times 10^5$. McKinnon et al. (2016).

Cells tens of km across overturn in $\sim 10^6$ yr, driven by present-day radiogenic heat in the rocky core.

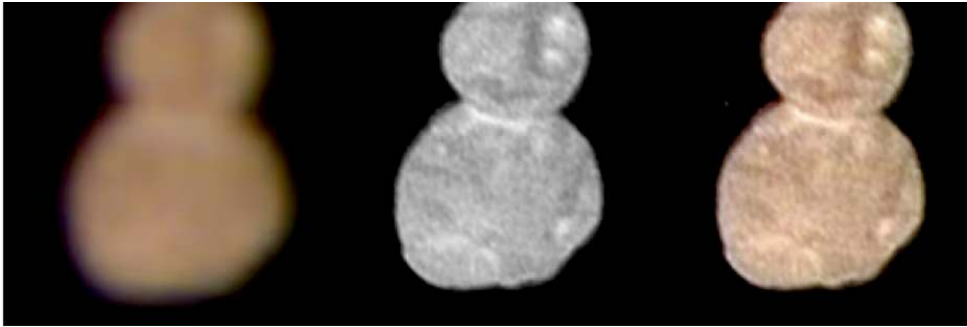


Pluto and Charon with their named features, New Horizons (2015). Stern et al. (2015).

Charon ($R = 606$ km, $\sim 1/8$ of Pluto's mass) puts the barycentre *outside* Pluto; the four small moons Styx, Nix, Kerberos and Hydra are likely fragments of an early collisional event.



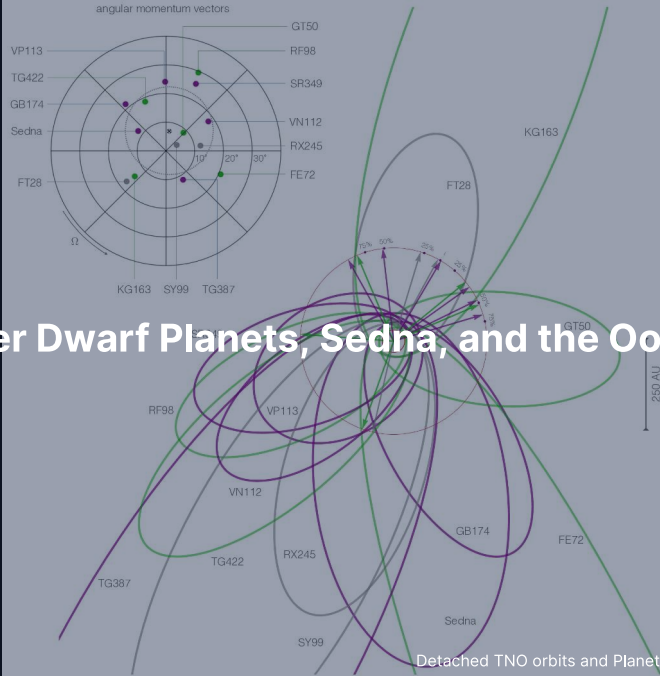
Charon from New Horizons: the dark reddish Mordor Macula of tholins from Pluto's escaping atmosphere, and equatorial extensional chasms. Credit: NASA/JHUAPL/SwRI, public domain.



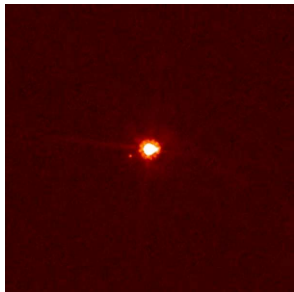
Arrokoth, a ~35 km **contact binary**, New Horizons flyby of 1 January 2019. Stern et al. (2019).

Two flat lobes joined at a narrow neck with no impact crater at the join: a merger at $\lesssim$ a few m s^{-1} , formation by streaming-instability gravitational collapse.

10. Other Dwarf Planets, Sedna, and the Oort Cloud

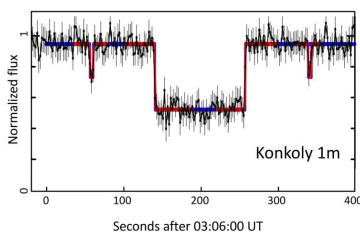


Eris, Haumea, Makemake



Eris + Dysnomia. More massive than Pluto.

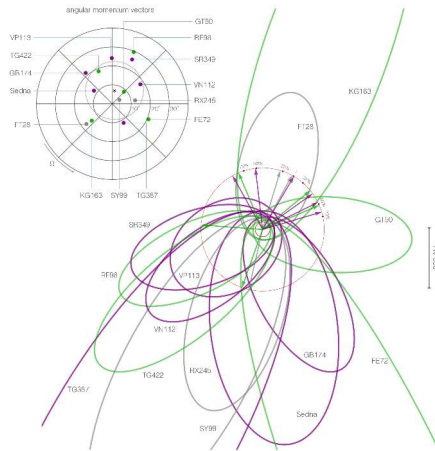
Plus dwarf-planet candidates: Gonggong, Quaoar, Orcus, Salacia, Sedna.



Haumea occultation: revealed a **ring**.

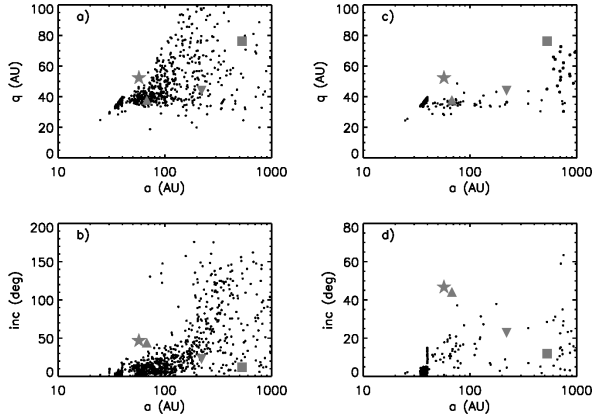


Makemake + small moon.
Methane-ice surface.



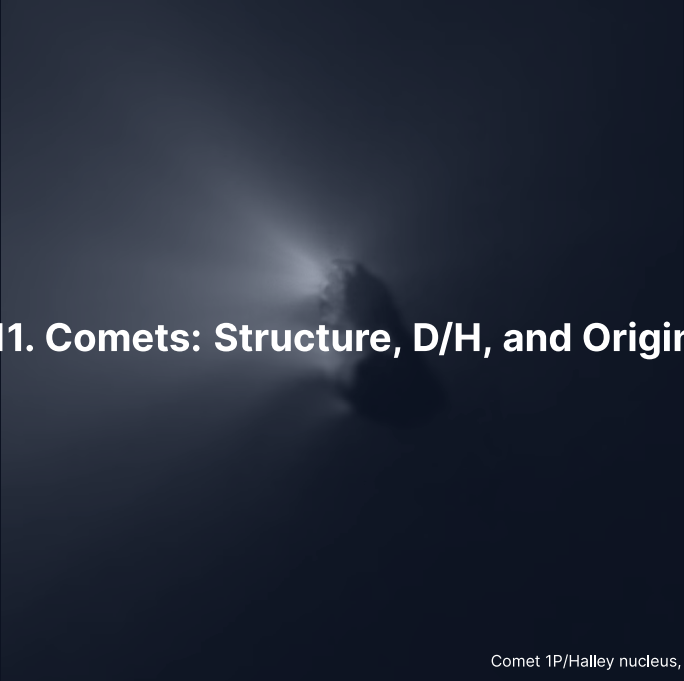
Orbits of the detached TNOs and the Planet 9 hypothesis. Batygin et al. (2019).

Sedna ($q = 76$ AU, $Q = 900$ AU, period 11,400 yr) is too far for Neptune and too close for present-day stars: an inner Oort cloud or birth-cluster relic; the orbital clustering motivates the **Planet 9** hypothesis.

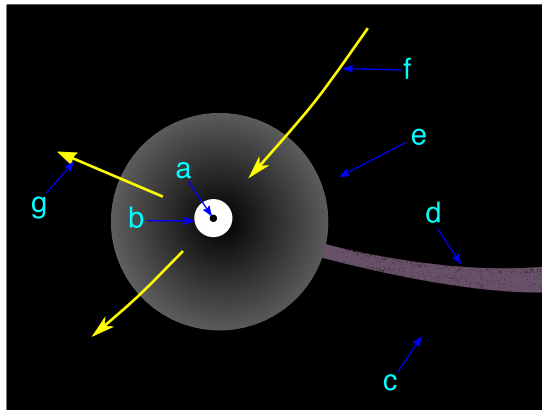


Two simulations of Oort cloud formation: perihelion q (top) and inclination (bottom) against semimajor axis a . Kaib & Quinn (2008).

2,000–50,000 AU, roughly spherical, $\sim 10^{11}$ – 10^{12} objects of ~ 1 – $10 M_{\oplus}$ in total, planetesimals scattered from the Jupiter-Neptune zone and lifted by the galactic tide and passing stars; inferred by Oort in 1950, **never directly observed**.



11. Comets: Structure, D/H, and Origin



Anatomy of an active comet: nucleus, coma, ion tail and dust tail. Credit: Wikimedia Commons, public domain.

Whipple's dirty snowball: a dark, porous **nucleus** of 1–30 km (67P: 533 kg m^{-3} , ~ 70% porosity) of ices, dust and organics. Inside ~ 5 AU water sublimation feeds a 10^4 – 10^6 km **coma**, a straight anti-sunward **ion tail** and a curved **dust tail**.



Halley's nucleus ($15 \times 8 \times 8$ km) with jets from small active patches, Giotto Halley Multicolour Camera, 13–14 March 1986. Credit: ESA/MPAe Lindau.

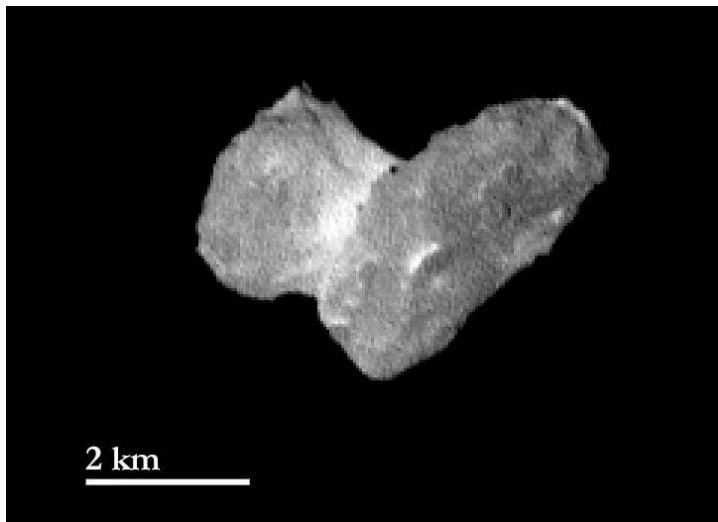
A 76-year retrograde orbit with $T_J \approx -0.6$: an Oort-cloud comet captured to a short period.

Short-Period vs Long-Period: The Tisserand Parameter

$$T_J = \frac{a_J}{a} + 2\sqrt{\frac{a}{a_J}(1-e^2)}\cos i, \quad a_J = 5.20 \text{ AU}$$

A conserved dynamical invariant that separates asteroids from Jupiter-family and long-period comets:

- **Asteroids** ($T_J > 3$): main-belt orbits decoupled from Jupiter encounters (e.g. Ceres, $T_J \approx 3.3$)
- **Jupiter-family comets** ($2 < T_J < 3$): Kuiper-belt-derived comets controlled by Jupiter (e.g. 67P, $T_J \approx 2.7$)
- **Long-period comets** ($T_J < 2$): Oort-cloud visitors on high-inclination orbits; Halley-type comets ($T_J \approx -0.6$) are Oort-derived but captured to short orbits (76 yr)



Bilobed nucleus of Jupiter-family comet 67P/Churyumov-Gerasimenko ($T_J \approx 2.7$), two lobes joined at a narrow neck by a gentle primordial merger, Rosetta OSIRIS. Credit: ESA/Rosetta/MPS for OSIRIS Team.



Comet 103P/Hartley 2 from EPOXI (2010), whose water D/H matches Earth's oceans. Credit: NASA/JPL-Caltech/UMD, public domain.

Earth's oceans have $D/H = 1.56 \times 10^{-4}$ (VSMOW); Hartley 2 matches, 67P is $\approx 3\times$ and long-period comets $\approx 2\times$ VSMOW. No comet class matches Earth: carbonaceous chondrites dominate Earth's water, comets supply at most a minor fraction.



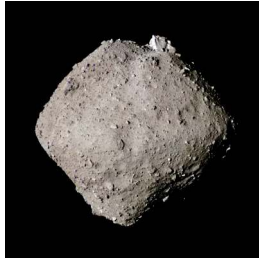
12. Recent Missions and Sample Return

2 km

Three Sample Returns: Itokawa, Ryugu, Bennu



Hayabusa, 2010: 1,500 grains.
Itokawa = LL chondrite parent.

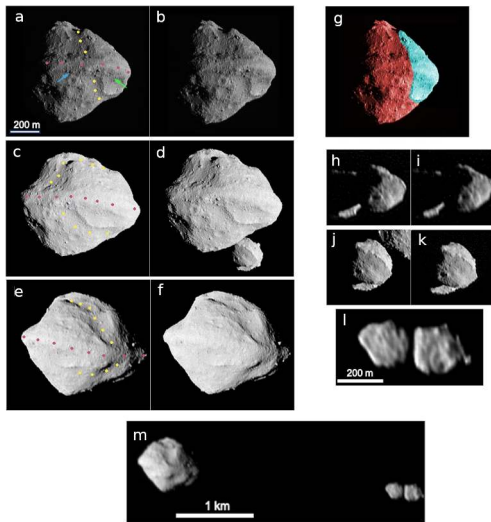


Hayabusa2, 2020: 5.4 g. Ryugu
≈ CI chondrite. Amino acids,
nucleobases.

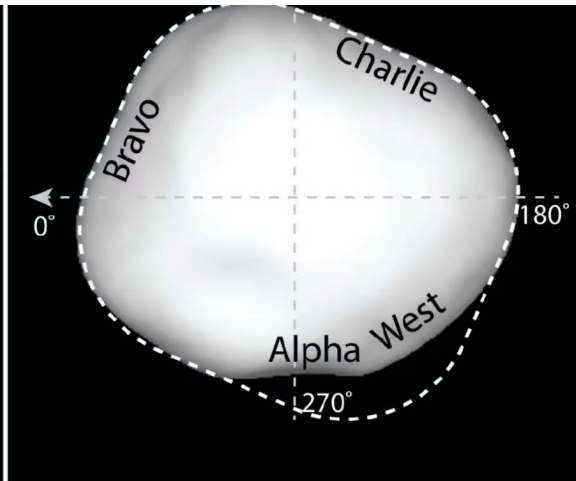
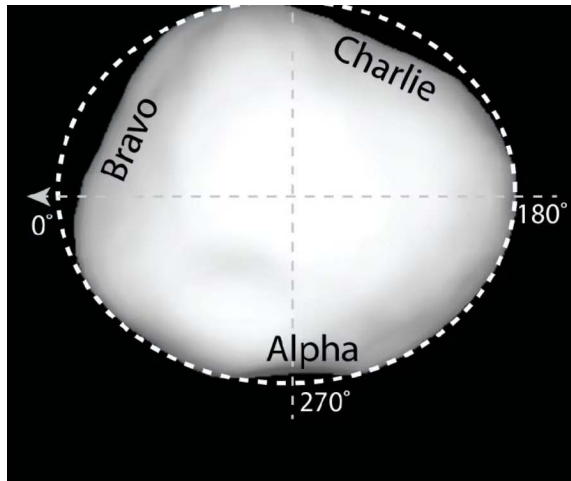


OSIRIS-REx, 2023: 121 g.
Bennu CM/CI-like, Mg
carbonate veins.

Each rock → specific parent body with a known orbit and full in-situ context.



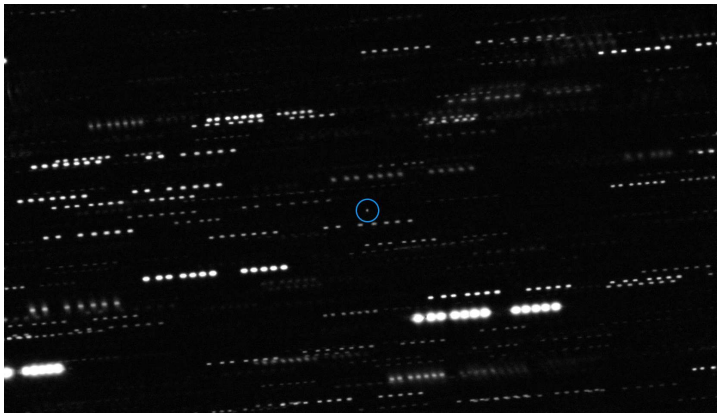
The Dinkinesh-Selam system from the Lucy flyby of 1 November 2023: the ~720 m main-belt asteroid with its equatorial ridge, and the moonlet Selam, a contact binary of two ~210 and 230 m lobes. Reproduced from Levison et al. (2024).



The shape of (16) Psyche from radar and adaptive-optics data, seen from above its south pole, with the regions Alpha, Bravo and Charlie where the body falls short of the best-fit ellipsoid. Reproduced from Shepard et al. (2021).

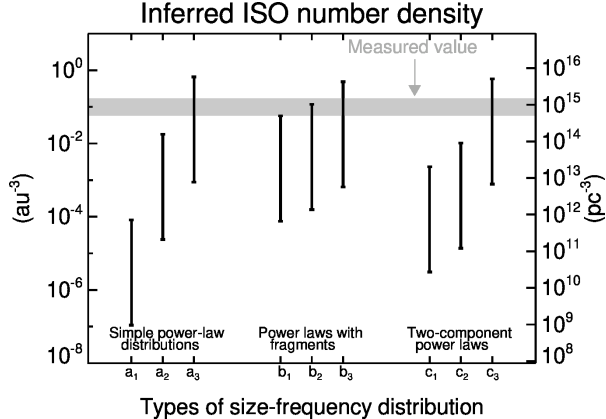


13. Interstellar Visitors



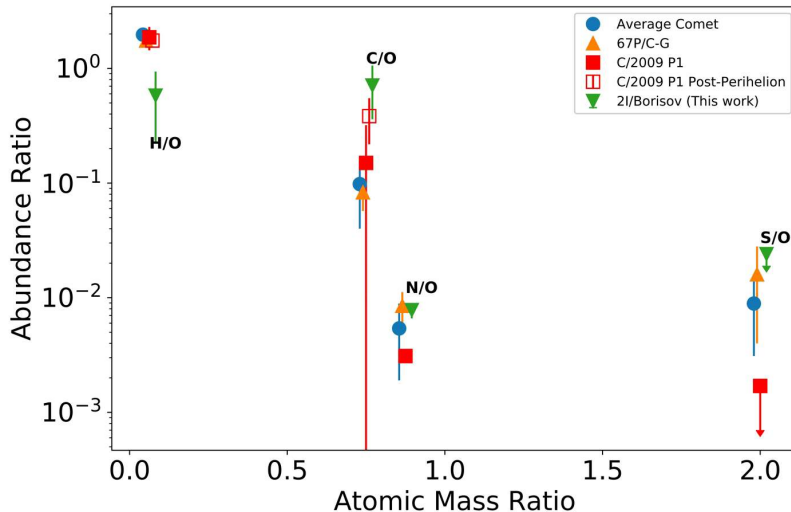
Deep image of 1I/'Oumuamua at the centre; the stars trail because the telescope tracked the target. Credit: ESO / K. Meech et al.

No detectable coma at the limit of stacked imaging, yet the outgoing trajectory shows a non-gravitational acceleration; its nature is still debated: H₂ ice fragment, N₂ ice, or tidal disruption shard.



Implied number density of interstellar objects from the single Pan-STARRS detection of 'Oumuamua. 'Oumuamua ISSI Team (2019).

One detection implies $\sim 10^4$ ISOs larger than 100 m within Neptune's orbit *at any moment*, each a sample of a different planetary system; the density exceeds the pre-discovery predictions by orders of magnitude.



Volatile composition in the coma of interstellar comet 2I/Borisov against solar-system comets: elevated C/O and depleted H/O point to formation beyond the CO snowline of its host system.

Credit: Bodewits et al. (2020), Fig. 3.

Summary

1. Small bodies are **formation fossils**; **Pb-Pb dating** anchors the solar system's absolute clock at 4567.30 ± 0.16 Myr
2. The **NC-CC dichotomy** separates inner and outer reservoirs for ~ 3–4 Myr; resonance gaps and Yarkovsky thermal recoil drive steady-state NEA delivery
3. Cometary D/H shows carbonaceous chondrites delivered most of Earth's water

Next: Lecture 13 turns to the exoplanet revolution.

Questions?

Next: **Lecture 13. Exoplanets**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`